Osculating-Cones Waverider Design Tool

Documentation of what is implemented so far

Theo Rulko

MIT AeroAstro

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Contents

1 Introduction **2**

2 Background **2**
2.1 Conical Shock Waves and the Flowfield . . . 2
2.2 Non-Dimensionalised Velocity . . . 2
2.3 Oblique Shock Relations . . . 2
2.4 Taylor-Maccoll Conical Flow . . . 3

3 Geometry Determination **4**
3.1 Parametric Trailing-Edge Curve . . . 4
3.2 Leading-Edge Projection onto the Shock Cone . . . 4
3.3 Streamline Tracing (Lower Surface Generation) . . . 4
3.4 Osculating-Plane Azimuth Angle . . . 4
3.5 Streamline ODE in Spherical Coordinates . . . 5
3.6 Base-Plane Intersection and Interpolation . . . 5
3.7 Cartesian Reconstruction . . . 5
3.8 Upper Surface . . . 5

4 Geometry Generation Algorithm Summary **5**

5 Inviscid aerodynamic forces **6**
5.1 Upper surface . . . 6
5.2 Lower surface . . . 6
5.3 Inviscid force integration and coefficients . . . 7

6 Viscous shear (skin friction) calculation **7**
6.1 Implementation . . . 8

7 Blunt leading-edge correction **9**
7.1 Stagnation-point sizing . . . 9
7.2 Modified-Newtonian force on a swept cylindrical leading edge . . . 9
7.3 Discretisation and integration . . . 10

8 References **10**

1 Introduction

This tool implements the **Osculating-Cones (OC) method** for generating waverider geometries at a prescribed design Mach number.

2 Background

2.1 Conical Shock Waves and the Flowfield

At the design Mach number M_1 the freestream encounters an axisymmetric conical shock at half-angle β (the *shock semi-angle*). Between the conical shock and the body cone the flow is described by the **Taylor-Maccoll** equations, which exploit the self-similar (conical) nature of the flowfield—all flow properties depend only on the polar angle θ measured from the cone axis.

Assumptions

- The shock is a *perfect* conical shock (no 3-D effects on the shock shape).
- The gas is *calorically perfect* (constant $\gamma = 1.4$), so real-gas and dissociation effects are neglected.
- The flowfield between shock and cone is *isentropic* and *irrotational*—valid because the shock is conical and hence of uniform strength.
- The flow is *steady* and *axisymmetric*.

2.2 Non-Dimensionalised Velocity

The Taylor-Maccoll formulation uses a velocity normalised by the maximum attainable speed $V_{\max}$, which corresponds to complete conversion of enthalpy to kinetic energy in an adiabatic process:

$$V_{\max} = \sqrt{\frac{2c_pT_0}{\gamma - 1} \cdot (\gamma - 1)} = \sqrt{\frac{2\gamma RT_0}{\gamma - 1}} \tag{1}$$

The non-dimensional speed $V' = V/V_{\max}$ can be related to the local Mach number through the energy equation:

$$\frac{1}{2}V'^2 + \frac{1}{\gamma - 1}\frac{a^2}{V_{\max}^2} = \frac{1}{2} \quad \Longrightarrow \quad V' = \frac{1}{\sqrt{\dfrac{2}{(\gamma - 1)M^2} + 1}} \tag{2}$$

where a is the local speed of sound and M the local Mach number.

2.3 Oblique Shock Relations

For a planar oblique shock at wave angle β with upstream Mach number M_1, the flow-deflection angle θ (theta) satisfies the **θ-β-M relation**:

$$\tan\theta = 2\cot\beta\,\frac{M_1^2\sin^2\beta - 1}{M_1^2(\gamma + \cos 2\beta) + 2} \tag{3}$$

The normal component of the upstream Mach number is

$$M_{n1} = M_1 \sin\beta \tag{4}$$

and the post-shock normal Mach number is given by the Rankine-Hugoniot normal-shock relation:

$$M_{n2}^2 = \frac{M_{n1}^2 + \dfrac{2}{\gamma - 1}}{\dfrac{2\gamma}{\gamma - 1} M_{n1}^2 - 1} \tag{5}$$

The total post-shock Mach number follows from the geometry of the shock:

$$M_2 = \frac{M_{n2}}{\sin(\beta - \theta)} \tag{6}$$

Initial Conditions for the Taylor-Maccoll ODE At the shock surface ($\theta = \beta$), the post-shock non-dimensional velocity components in spherical coordinates (r, θ) are:

$$V'_{r,i} = V' \cos(\beta - \theta) \tag{7}$$

$$V'_{\theta,i} = -V' \sin(\beta - \theta) \tag{8}$$

where V' is computed from equation (2) evaluated at M_2. These form the initial value vector $\mathbf{S}_0 = (V'_{r,i}, V'_{\theta,i})$ for the ODE integration.

2.4 Taylor-Maccoll Conical Flow

For *steady, irrotational, isentropic* flow about an axisymmetric cone with a self-similar (conical) structure, the velocity field in spherical coordinates (r, θ, ϕ) reduces to $\mathbf{V} = V_r(\theta)\,\hat{r} + V_\theta(\theta)\,\hat{\theta}$. Substituting into the full-potential equation and using the energy constraint yields the **Taylor-Maccoll equation** in non-dimensional form (Anderson, *Modern Compressible Flow*, Eq. 10.15):

$$\frac{\gamma - 1}{2}\left(1 - V_r'^2 - V_\theta'^2\right)\left(2V'_r + \frac{V'_\theta}{\tan\theta}\right) - V'_r V_\theta'^2 = V'_\theta \left(V'_r V'_\theta - \frac{\gamma - 1}{2}(1 - V_r'^2 - V_\theta'^2)\cot\theta\right)' \tag{9}$$

Introducing $A = V'_r$ and $A' = \mathrm{d}V'_r/\mathrm{d}\theta$ (so $V'_\theta = A'$ by the irrotationality condition), the equation is rewritten as an explicit second-order ODE for A:

$$\boxed{A'' = \frac{A\,A'^2 - \dfrac{\gamma - 1}{2}(1 - A^2 - A'^2)\left(2A + \dfrac{A'}{\tan\theta}\right)}{\dfrac{\gamma - 1}{2}(1 - A^2 - A'^2) - A'^2}} \tag{10}$$

This is split into the first-order system:

$$\frac{\mathrm{d}}{\mathrm{d}\theta}\begin{pmatrix} A \\ A' \end{pmatrix} = \begin{pmatrix} A' \\ A'' \end{pmatrix} \tag{11}$$

Integration proceeds *inward* from the shock surface ($\theta = \beta$, where initial conditions are set by the oblique-shock solution) toward the cone axis ($\theta \to 0$). The body cone is located at the angle θ_c where $V'_\theta = A' = 0$, i.e. the streamlines are parallel to the cone surface. This zero crossing is the natural **terminal event**:

$$\text{event: } A'(\theta_c) = 0 \tag{12}$$

The integration interval is $[\beta,\ 10^{-8}]$ (nearly zero, to avoid the axis singularity).

The ODE is solved with `scipy.integrate.solve_ivp` using the **Runge-Kutta 4(5) (RK45)** method with tight tolerances (rtol $= 10^{-8}$, atol $= 10^{-10}$) to capture the near-axis behaviour accurately. The event (12) is registered as a terminal event, halting integration once the cone surface is found.

3 Geometry Determination

3.1 Parametric Trailing-Edge Curve

The trailing edge (TE) lies in the *base plane* $x = L$ (where L is the maximum axial length of the vehicle). In the (y, z) cross-section, the TE is defined by a **quartic polynomial**:

$$z_{\mathrm{TE}}(y) = a + b\,y^2 + c\,y^4 \tag{13}$$

The coefficients a, b, c are derived from three user-specified geometric constraints:

$$R_1 = 0.2\,R_s \qquad \text{(inboard semi-height at centreline)} \tag{14}$$

$$W_2 = 0.8\,R_s \qquad \text{(half-span to outer tangency point)} \tag{15}$$

$$R_s = L\tan\beta \qquad \text{(shock radius at base plane)} \tag{16}$$

Explicitly:

$$a = -R_1, \quad b = \frac{2\left(R_1 - \sqrt{R_s^2 - W_2^2}\right)}{W_2^2}, \quad c = \frac{\sqrt{R_s^2 - W_2^2} - R_1}{W_2^4} \tag{17}$$

The lateral bound y_{up} of the TE curve is the y-value at which the curve meets the shock circle:

$$\sqrt{y_{\mathrm{up}}^2 + z_{\mathrm{TE}}(y_{\mathrm{up}})^2} = R_s \tag{18}$$

This nonlinear equation is solved with `scipy.optimize.fsolve`.

3.2 Leading-Edge Projection onto the Shock Cone

The leading edge (LE) is obtained by projecting every point on the TE curve radially *inward* along the shock cone surface. For a point $(y, z) = (y_{\mathrm{TE}}, z_{\mathrm{TE}})$ in the base plane, the projection maintains the same azimuthal direction while scaling the axial coordinate to keep the point on the cone. The radial distance from the cone axis is $\rho = \sqrt{y^2 + z^2}$; because the shock cone satisfies $\rho = x\tan\beta$, the axial coordinate of the LE is:

$$X_{\mathrm{LE}} = L\left(1 - \frac{R_s - \rho}{R_s}\right) = \frac{L\,\rho}{R_s} \tag{19}$$

with $Y_{\mathrm{LE}} = y_{\mathrm{TE}}$ and $Z_{\mathrm{LE}} = z_{\mathrm{TE}}$.

3.3 Streamline Tracing (Lower Surface Generation)

The full 3-D waverider lower surface is generated by tracing streamlines in the conical flowfield. Because the shock is a cone, each streamline lies in an **osculating plane**—a half-plane emanating from the cone axis that contains the local leading-edge tangent. Within each osculating plane, the 2-D conical flow solution (Taylor-Maccoll) is applied exactly; the planes are simply rotated in azimuth ϕ to span the full TE curve. This is the **Osculating Cones** approximation (Sobieczky *et al.*, 1990; Jones *et al.*, 1995).

3.4 Osculating-Plane Azimuth Angle

For the k-th LE seed point with Cartesian coordinates (X_k, Y_k, Z_k) at the shock, the azimuthal (roll) angle of the osculating plane is:

$$\phi_k = -\arctan\left(\frac{Y_k}{Z_k}\right) \tag{20}$$

3.5 Streamline ODE in Spherical Coordinates

A streamline in the conical flowfield satisfies:

$$\frac{\mathrm{d}r}{\mathrm{d}\theta} = r\,\frac{V_r'(\theta)}{V_\theta'(\theta)} \tag{21}$$

where $V_r'(\theta)$ and $V_\theta'(\theta)$ are provided by the Taylor-Maccoll solution.

Equation (21) is discretised with the explicit **Euler** method. Starting from (r_0, θ_0) at the shock surface, successive points are computed as:

$$r_{i+1} = r_i + \frac{V_r'(\theta_i)}{V_\theta'(\theta_i)}\, r_i\, \Delta\theta \tag{22}$$

where $\Delta\theta = \theta_{i+1} - \theta_i < 0$ (integration proceeds toward the axis). The step size is inherited from the Taylor-Maccoll solution grid (1000 uniform steps between θ_0 and 10^{-8} rad).

3.6 Base-Plane Intersection and Interpolation

Integration stops when the streamline crosses the base plane $x = L$, i.e. when $r_i \cos\theta_i < L \leq r_{i+1}\cos\theta_{i+1}$. At that crossing a linear interpolation gives the precise intersection point. In the osculating plane the interpolated z-coordinate (perpendicular to x within the osculating plane) is:

$$z^* = \cos\phi \left[\frac{r_{i+1}\sin\theta_{i+1} - r_i \sin\theta_i}{r_{i+1}\cos\theta_{i+1} - r_i\cos\theta_i}\, L + r_i \sin\theta_i - r_i\cos\theta_i\, \frac{r_{i+1}\sin\theta_{i+1} - r_i\sin\theta_i}{r_{i+1}\cos\theta_{i+1} - r_i\cos\theta_i} \right] \tag{23}$$

From z^* the intersection spherical coordinates are recovered:

$$r^* = \sqrt{(z^*/\cos\phi)^2 + L^2} \tag{24}$$

$$\theta^* = \arctan\left(\frac{z^*/\cos\phi}{L}\right) \tag{25}$$

3.7 Cartesian Reconstruction

After integration, each streamline point (r_j, θ_j) is transformed back to Cartesian coordinates by rotating around the x-axis by ϕ:

$$x_j = r_j \cos\theta_j \tag{26}$$

$$y_j = \quad r_j \sin\theta_j \sin\phi \tag{27}$$

$$z_j = -r_j \sin\theta_j \cos\phi \tag{28}$$

Mirrored streamlines (opposite side of the vehicle) are generated by negating y_j.

3.8 Upper Surface

The upper surface is approximated as a ruled surface swept from LE break points straight to the TE at constant (y, z)—a flat top that sees only the freestream. This is consistent with the waverider design philosophy of generating lift only from the lower surface.

4 Geometry Generation Algorithm Summary

Step 1. Input: read M_1, γ, β, trailing edge description

Step 2. Oblique shock: solve equations (3)–(6) to obtain M_2, θ; form initial conditions $(V_{r,i}', V_{\theta,i}')$ via equations (7)–(8).

Step 3. Taylor-Maccoll: integrate equation (10) from $\theta = \beta$ inward with RK45 until $V'_\theta = 0$; record body-cone angle θ_c.

Step 4. Trailing edge: evaluate quartic TE curve (equation 13) over $y \in [-y_{\text{up}}, y_{\text{up}}]$; compute break-point array in (x, y, z).

Step 5. Leading edge: project TE points onto shock cone via equation (19).

Step 6. Streamline tracing: for each of N_l seed points on the LE:

a. compute osculating-plane azimuth ϕ (eq. 20),
b. re-integrate Taylor-Maccoll on uniform θ-grid,
c. Euler-march equation (22) until base-plane crossing,
d. interpolate exact intersection (eq. 23),
e. convert to Cartesian coordinates (eq. 28).

Step 7. B-spline: fit smooth curve through base-plane endpoints.

Step 8. Upper surface: ruled lines from LE break points to TE.

5 Inviscid aerodynamic forces

5.1 Upper surface

The upper surface of the waverider is a ruled flat top that sees only the undisturbed freestream. Thus, conditions at the boundary-layer edge are simply the freestream conditions:

$$u_e = V_\infty, \quad T_e = T_\infty, \quad p_e = p_\infty, \quad \rho_e = \rho_\infty, \quad M_e = M_1. \tag{29}$$

Therefore, the pressure coefficient over the upper surface is zero, and it contributes no pressure drag (but it will contribute to the viscous drag, which is calculated in the next section).

5.2 Lower surface

Because the lower surface is a stream surface of the Taylor–Maccoll cone, each streamline lies in an osculating plane and its boundary-layer-edge properties are functions of the polar angle θ from the cone axis. For each Cartesian point (x, y, z) on a streamline we compute $\theta = \arctan\!\left(\sqrt{y^2 + z^2}/x\right)$ and look up $V'(\theta)^2 = V_r'^2 + V_\theta'^2$ from a single Taylor–Maccoll solution. Total temperature and post-shock total pressure are constant in the conical region; this gives, with freestream T_∞ and p_∞:

$$T_0 = T_\infty\left(1 + \tfrac{\gamma-1}{2}M_1^2\right), \quad V_{\max} = \sqrt{2\gamma R T_0/(\gamma-1)}, \tag{30}$$

$$u_e = V_{\max}\, V'(\theta), \qquad T_e = T_0\left(1 - V'^2(\theta)\right), \qquad M_e = u_e/\sqrt{\gamma R T_e}, \tag{31}$$

$$p_e = p_{02}\left(1 - V'^2(\theta)\right)^{\gamma/(\gamma-1)}, \qquad \rho_e = p_e/(RT_e), \tag{32}$$

where p_{02} is the post-shock stagnation pressure. In the implementation this is obtained from the oblique-shock relations using the normal component $M_{n1} = M_1 \sin\beta$:

$$\frac{p_2}{p_1} = \frac{2\gamma M_{n1}^2 - (\gamma-1)}{\gamma+1}, \qquad \frac{p_{02}}{p_1} = \frac{p_2}{p_1}\left(1 + \tfrac{\gamma-1}{2}M_2^2\right)^{\gamma/(\gamma-1)}. \tag{33}$$

The pressure coefficient follows from $q_\infty = \frac{\gamma}{2}M_1^2 p_1$:

$$C_p = \frac{p_e/p_1 - 1}{\frac{\gamma}{2}M_1^2} = \frac{\left(\dfrac{p_{02}}{p_1}\right)(1 - V'^2)^{\gamma/(\gamma-1)} - 1}{\frac{\gamma}{2}M_1^2}. \tag{34}$$

5.3 Inviscid force integration and coefficients

Only the lower surface contributes to inviscid pressure forces (the upper surface has $C_p = 0$). The lower surface is panelized into triangles with area $\mathrm{d}A$ and outward unit normal $\hat{\mathbf{n}}$. The pressure force on each panel is integrated in coefficient form as

$$\mathrm{d}\mathbf{C}_F = -C_p\,\hat{\mathbf{n}}\,\frac{\mathrm{d}A}{S_{\mathrm{ref}}}, \tag{35}$$

so the total force coefficient vector is $\mathbf{C}_F = \sum \mathrm{d}\mathbf{C}_F$. The reference area is the planform (projection) of the lower surface onto the x-y plane,

$$S_{\mathrm{ref}} = \sum \left| \frac{1}{2}\left[(x_1 - x_0)(y_2 - y_0) - (y_1 - y_0)(x_2 - x_0)\right] \right|, \tag{36}$$

using the projected triangle vertices (x_i, y_i). With the body traveling in $+x$ and lift positive in $+z$, the coefficient components are reported as $C_L = C_{F,z}$, $C_D = C_{F,x}$, and $L/D = C_L/C_D$.

6 Viscous shear (skin friction) calculation

To estimate viscous drag on the waverider surface, the laminar compressible integral boundary-layer method of Walz was used, following the implementation reported by Bowcutt in his viscous-optimized waverider design paper. We obtain the boundary layer edge conditions $\{u_e, M_e, \rho_e, T_e\}$ from the external inviscid flow (Taylor–Macoll) and march the boundary layer downstream by solving a coupled pair of first-order ordinary differential equations along a streamline on the boundary-layer edge (the flow is assumed to be locally 2D). The model accounts for compressibility, wall-temperature effects, and streamwise pressure-gradient effects through the prescribed edge flow. It assumes a steady, continuum, laminar, thin, attached boundary layer with no flow separation or transition to turbulence. It assumes an isothermal wall at T_w; radiation equilibrium is not modelled.

The two governing equations are the integral momentum equation and the integral mechanical-energy equation, where s is a coordinate along the boundary-layer-edge streamline, and y is the coordinate normal to the surface:

$$\frac{\mathrm{d}Z}{\mathrm{d}s} + \frac{1}{u_e}\frac{\mathrm{d}u_e}{\mathrm{d}s}\,F_1\,Z - F_2 = 0, \qquad \frac{\mathrm{d}W}{\mathrm{d}s} + \frac{1}{u_e}\frac{\mathrm{d}u_e}{\mathrm{d}s}\,F_3\,W - \frac{F_4}{Z} = 0, \tag{37}$$

where Z, W and the boundary layer thicknesses δ_i are defined as:

$$Z = \delta_2\left(\frac{\rho_e u_e \delta_2}{\mu_w}\right) = \frac{\rho_e u_e \delta_2^2}{\mu_w}, \qquad W = \frac{\delta_3}{\delta_2}, \tag{38}$$

$$\delta_1 = \int_0^\delta \left(1 - \frac{\rho u}{\rho_e u_e}\right)\mathrm{d}y, \qquad \delta_2 = \int_0^\delta \frac{\rho u}{\rho_e u_e}\left(1 - \frac{u}{u_e}\right)\mathrm{d}y, \qquad \delta_3 = \int_0^\delta \frac{\rho u}{\rho_e u_e}\left(1 - \frac{u^2}{u_e^2}\right)\mathrm{d}y, \tag{39}$$

were $(\cdot)_w$ denotes evaluation at the wall. The closures in these equations (copied from the textbook) are:

$$F_1 = 3 + 2H - M_e^2 + n\,\frac{\dfrac{1}{\mu_w}\dfrac{\mathrm{d}\mu_w}{\mathrm{d}s}}{\dfrac{1}{u_e}\dfrac{\mathrm{d}u_e}{\mathrm{d}s}}, \qquad n = \begin{cases} 0, & T_w = \text{constant}, \\ 1, & T_w \neq \text{constant}, \end{cases} \tag{40}$$

$$F_2 = \frac{2a}{b}, \qquad F_3 = 1 - H + r(\gamma - 1)M_e^2\left(1 - \frac{\widetilde{\theta}}{W}\right), \qquad F_4 = \frac{2\beta - aW}{b}, \tag{41}$$

and

$$H = \frac{\delta_1}{\delta_2} = b\,H_{12} + r\,\frac{\gamma-1}{2}\,M_e^2\left(W - \widetilde{\theta}\right), \tag{42}$$

$$a = 1.7261\,(W^* - 1.515)^{0.7158}, \qquad b = \frac{\delta_2^{(u)}}{\delta_2} = 1 + r\,\frac{\gamma-1}{2}\,M_e^2\left(W - \widetilde{\theta}\right)(2 - W), \tag{43}$$

$$r = \sqrt{\mathrm{Pr}}, \qquad \widetilde{\theta} = \frac{T_{aw}(s) - T_w(s)}{T_{aw}(s) - T_e(s)}, \qquad \beta = \beta^{(u)}\,\chi, \tag{44}$$

and

$$H_{12} = 4.0306 - 4.2845\,(W^* - 1.515)^{0.3886}, \qquad T_{aw} = T_e + \frac{r\,u_e^2}{2c_p}, \quad W^* = \frac{\delta_3^{(u)}}{\delta_2^{(u)}} = \frac{W}{\psi}, \tag{45}$$

$$\psi = 1 + \frac{(\psi_{12} - 1)M_e^2}{M_e^2 + \psi_{12}^{-1}/\psi_0}, \qquad \psi_{12} = \frac{2 - \delta_1^{(u)}/\delta}{W^*}\,\widetilde{\theta} + \frac{1 - \delta_1^{(u)}/\delta}{W^* g}\,(1 - \widetilde{\theta}), \tag{46}$$

$$\psi_0 = 0.0144\,(2 - W^*)\,(2 - \widetilde{\theta})^{0.8}, \qquad \frac{\delta_1^{(u)}}{\delta} = 0.420 - (W^* - 1.515)^{0.424\,W^*}, \tag{47}$$

$$g = 0.324 + 0.336\,(W^* - 1.515)^{0.555}, \qquad \beta^{(u)} = 0.1564 + 2.1921\,(W^* - 1.515)^{1.70}, \tag{48}$$

$$\chi = \left\{1 + r\,\frac{\gamma-1}{2}\,M_e^2\left[1.16\,W^* - 1.072 - \widetilde{\theta}(2W^* - 2.581)\right]\right\}^{0.7}\left\{1 + r\,\frac{\gamma-1}{2}\,M_e^2(1 - \widetilde{\theta})\right\}^{-0.7}. \tag{49}$$

Wall viscosity $\mu_w(T_w)$ is given by Sutherland's law,

$$\mu(T) = \mu_0\left(\frac{T}{T_0^\mu}\right)^{3/2}\frac{T_0^\mu + S}{T + S}, \quad \mu_0 = 1.716\times10^{-5}\ \mathrm{Pa\,s},\ T_0^\mu = 273.15\ \mathrm{K},\ S = 110.4\ \mathrm{K}. \tag{50}$$

6.1 Implementation

Initial conditions and self-similar start

At the leading edge $s = 0$ the boundary layer has zero thickness, so $Z \to 0$ is singular for the W-equation. We start the integration at a small $s_0 > 0$ from the local self-similar (zero pressure-gradient) state. Setting $\mathrm{d}W/\mathrm{d}s = 0$ in the W-equation with $\mathrm{d}u_e/\mathrm{d}s = 0$ gives the algebraic fixed point $W = 2\beta/a$. With Eq. (45) this is iterated together with the $W \leftrightarrow W^*$ inversion until self-consistent (the incompressible Blasius value $W^* \approx 1.572$ is used as the seed). The matching Z is then $Z(s_0) = (2a/b)\,s_0$, which is the linear Blasius-like growth.

Numerical solution

For each streamline the integration uses the following pipeline:

1. Sample the streamline to a uniform grid in arc length s.
2. Tabulate $\left(u_e, T_e, M_e, \rho_e, \mu_w, \widetilde{\theta}\right)$ at each station; build linear interpolants to evaluate them between stations.
3. Integrate the coupled system $\left(Z(s), W(s)\right)$ with `scipy.integrate.solve_ivp`, RK45, rtol $= 10^{-7}$, atol $= 10^{-10}$.
4. At every RHS evaluation, given the current compressible W and the local edge state, solve the implicit relation (45) $W = W^*\,\psi(W^*)$ for W^* with `scipy.optimize.brentq` on $W^* \in (1.515, 1.99)$, then evaluate a, b, H_{12}, H, $\beta^{(u)}$, χ, β and the four coefficients F_1, F_2, F_3, F_4.

The momentum thickness, wall shear, and skin-friction coefficient are recovered post-integration by:

$$\delta_2 = \sqrt{\frac{Z\,\mu_w}{\rho_e u_e}}, \qquad \tau_w = \frac{a\,\mu_w u_e}{\delta_2}, \qquad c_f = \frac{2\tau_w}{\rho_e u_e^2}. \tag{51}$$

Skin-friction drag integration

On both surfaces, each streamline represents a strip of the wetted surface; strip width $w(s)$ is approximated by the half-distance to each neighbouring streamline at the same fractional arc length (with a single-neighbour doubling at the outermost streamlines). The streamwise friction force on the strip is

$$F_{x,k} \;=\; \int_0^{s_{\mathrm{TE}}} \tau_w(s)\,\big(\mathrm{d}x/\mathrm{d}s\big)\,w(s)\,\mathrm{d}s, \tag{52}$$

evaluated by the trapezoidal rule. The total friction-drag coefficient sums the contributions from both surfaces, each doubled to account for the mirrored half:

$$C_{D_f} \;=\; \frac{2}{q_\infty S_{\mathrm{ref}}}\left(\sum_{k=1}^{N_l} F_{x,k}^{\mathrm{lower}} + \sum_{k=1}^{N_{up}} F_{x,k}^{\mathrm{upper}}\right), \qquad C_{D_f} = C_{D_f}^{\mathrm{lower}} + C_{D_f}^{\mathrm{upper}}, \tag{53}$$

with S_{ref} the planform projection of the lower surface.

7 Blunt leading-edge correction

The osculating-cones construction produces an aerodynamically sharp leading edge (LE), which is unphysical: stagnation-point convective heating scales as $1/\sqrt{R_n}$ and would be unbounded. A finite blunting radius R_n is required to keep the surface temperature below a chosen material limit T_{allow}. The blunt LE is treated as a circular cylinder of radius R_n swept along the streamline-derived LE curve, and its inviscid pressure force is added to the sharp-LE result via the modified-Newtonian theory.

7.1 Stagnation-point sizing

Convective heating at the stagnation point is estimated from the Sutton–Graves correlation for air,

$$\dot{q}_{\mathrm{conv}} = k_{\mathrm{SG}}\sqrt{\frac{\rho_\infty}{R_n}}\,V_\infty^3, \qquad k_{\mathrm{SG}} = 1.83\times10^{-4}\ \mathrm{W\,s^3\,kg^{-1/2}\,m^{-7/2}}, \tag{54}$$

and balanced against radiative equilibrium with emissivity ε, Stefan–Boltzmann constant σ, and background temperature T_{bg}:

$$\dot{q}_{\mathrm{rad}} = \varepsilon\sigma\big(T_w^4 - T_{\mathrm{bg}}^4\big)\,. \tag{55}$$

Setting $\dot{q}_{\mathrm{conv}} = \dot{q}_{\mathrm{rad}}$ at $T_w = T_{\mathrm{allow}}$ and inverting for the radius gives

$$\boxed{R_{n,\min} = \rho_\infty\left(\frac{k_{\mathrm{SG}}\,V_\infty^3}{\varepsilon\sigma\Big(T_{\mathrm{allow}}^4 - T_{\mathrm{bg}}^4\Big)}\right)^2.} \tag{56}$$

7.2 Modified-Newtonian force on a swept cylindrical leading edge

The discretised leading-edge curve is processed segment by segment. For a segment with unit tangent $\hat{\mathbf{t}}$ and freestream direction $\hat{\mathbf{u}}_\infty = \hat{\mathbf{x}}$, the local sweep angle Λ is defined by

$$\sin\Lambda = \hat{\mathbf{t}}\cdot\hat{\mathbf{u}}_\infty, \qquad \hat{\mathbf{u}}_\perp = \frac{\hat{\mathbf{u}}_\infty - \sin\Lambda\,\hat{\mathbf{t}}}{\cos\Lambda}, \tag{57}$$

where $\hat{\mathbf{u}}_\perp$ is the unit vector along the freestream component perpendicular to the LE. The Mach number normal to the leading edge is $M_\perp = M_1 \cos\Lambda$, and the modified-Newtonian peak pressure coefficient (evaluated at $M_\perp$) follows from the normal-shock stagnation-pressure construction

$$C_{p,\max}(M_\perp) = \frac{p_{02}/p_1 - 1}{\frac{1}{2}\gamma M_\perp^2}, \tag{58}$$

where p_{02}/p_1 is built from the standard normal-shock and isentropic relations applied to $M_\perp$.

In the plane perpendicular to $\hat{\mathbf{t}}$ the cylinder cross-section is a circle of radius R_n, parameterised by an angle ϕ measured from the stagnation line. The modified-Newtonian distribution is

$$C_p(\phi) = C_{p,\max}(M_\perp)\,\cos^2\Lambda\,\cos^2\phi, \qquad |\phi| \le \pi/2 \text{ (windward)}, \tag{59}$$

and zero on the leeward side. Integrating the pressure load $-C_p\, q_\infty\, R_n\, \hat{\mathbf{n}}\, \mathrm{d}\phi\, \mathrm{d}s$ over the windward half exploits $\int_{-\pi/2}^{\pi/2} \cos^3\phi\, \mathrm{d}\phi = 4/3$ and gives a closed-form resultant force per unit LE arc length,

$$\boxed{\frac{\mathrm{d}\mathbf{F}}{\mathrm{d}s} = \tfrac{4}{3}\, q_\infty\, R_n\, C_{p,\max}(M_\perp)\,\cos^2\Lambda\,\hat{\mathbf{u}}_\perp.} \tag{60}$$

The transverse component (in the $\hat{\mathbf{t}} \times \hat{\mathbf{u}}_\perp$ direction) vanishes by symmetry of $\cos^2\phi$ about $\phi = 0$.

7.3 Discretisation and integration

The leading-edge curve $\{(X_k, Y_k, Z_k)\}_{k=1}^N$ from §3.2 is differenced to obtain segment tangents $\hat{\mathbf{t}}_k$ and arc lengths Δs_k. Equation (60) is evaluated at every segment and summed,

$$\mathbf{F}_{\mathrm{LE}} \;=\; \sum_k \tfrac{4}{3}\, q_\infty\, R_n\, C_{p,\max}(M_{\perp,k})\,\cos^2\Lambda_k\,\hat{\mathbf{u}}_{\perp,k}\,\Delta s_k. \tag{61}$$

Because the discretised LE spans the full vehicle ($y \in [-y_{\mathrm{up}}, +y_{\mathrm{up}}]$) no symmetry doubling is required. The resultant lift- and drag-coefficient increments are

$$\Delta C_L = \frac{F_{\mathrm{LE},z}}{q_\infty S_{\mathrm{ref}}}, \qquad \Delta C_D = \frac{F_{\mathrm{LE},x}}{q_\infty S_{\mathrm{ref}}}, \tag{62}$$

which are added to the sharp-LE inviscid coefficients of §4 and the viscous skin-friction drag of §5 to obtain the total aerodynamic coefficients. Segments for which $M_\perp \le 1$ (i.e. $\cos\Lambda < 1/M_1$) are assigned $C_{p,\max} = 0$, since the modified-Newtonian construction relies on a detached normal shock.

8 References

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